## Create a tomcat container on 8080 and deploy sample application in tomcat.

```
[root@ip-10-0-1-114 ~]# dnf update -y
Amazon Linux 2023 Kernel Livepatch repository                                   949 kB/s |  89 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
```

```
[root@ip-10-0-1-114 ~]# dnf install docker -y
Last metadata expiration check: 0:00:33 ago on Wed Oct 14 10:19:22 2026.
Dependencies resolved.
================================================================================================================
 Package                          Architecture          Version                          Repository
================================================================================================================
Installing:
 docker                           x86_64                25.0.16-1.amzn2023.0.5           amazonlinux
Installing dependencies:
 container-selinux                noarch                4:2.245.0-1.amzn2023             amazonlinux
 containerd                       x86_64                2.3.5-1.amzn2023.0.1             amazonlinux
 iptables-libs                    x86_64                1.8.8-3.amzn2023.0.2             amazonlinux
```

dnf install docker – y  ---> to install docker

```
[root@ip-10-0-1-114 ~]# systemctl is-active docker
active
[root@ip-10-0-1-114 ~]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-10-0-1-114 ~]# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

```
[root@ip-10-0-1-114 ~]# docker pull tomcat:9.0
9.0: Pulling from library/tomcat
4e07a0f12b2c: Pull complete
06ad70e463aa: Pull complete
5dec61325458: Pull complete
f39241fdc590: Pull complete
4f4fb700ef54: Pull complete
130739ca07b1: Pull complete
954939152883: Pull complete
6db72904ff0f: Pull complete
Digest: sha256:cd946066142aaa3aa79abe69181ede8b3efdf0fdce2a1847826307d108254a18
Status: Downloaded newer image for tomcat:9.0
docker.io/library/tomcat:9.0
```

Pull the tomcat using ----> docker pull tomcat:9.0

```
[root@ip-10-0-1-114 ~]# docker run -d --name tomcat-8080 -p 8080:8080 tomcat:9.0
c10e5635a8c67b691d5a7c83037d00b0d1ebb36ad915665d8e46ad1e4179fd39
[root@ip-10-0-1-114 ~]# docker ps
CONTAINER ID   IMAGE        COMMAND             CREATED          STATUS          PORTS                                       NAMES
c10e5635a8c6   tomcat:9.0   "catalina.sh run"   12 seconds ago   Up 11 seconds   0.0.0.0:8080->8080/tcp, :::8080->8080/tcp   tomcat-8080
[root@ip-10-0-1-114 ~]# docker logs tomcat-8080
NOTE: Picked up JDK_JAVA_OPTIONS:  --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.lang.invoke=ALL-UNNAMED --add-open
s=java.base/java.lang.reflect=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=
java.base/java.util.concurrent=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED
14-Oct-2026 10:27:14.557 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Server version name:   Apache Tomcat/9.0.123
14-Oct-2026 10:27:14.564 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Server built:          Oct 6 2026 18:59:20 UTC
14-Oct-2026 10:27:14.565 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Server version number: 9.0.123.0
14-Oct-2026 10:27:14.565 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log OS Name:               Linux
14-Oct-2026 10:27:14.565 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log OS Version:            6.18.54-121.166.amzn2023.x
86_64
```

Creating a container

Create a sample application using -----> mkdir -p /tmp/sample-app

mkdir creates a directory.

-p creates parent directories if needed and avoids an error if the directory already exists.

/tmp/sample-app is where we'll create our sample webpage.

```
[root@ip-10-0-1-114 ~]# mkdir -p /tmp/sample-app
<ker Tomcat</h1><p>Task 1 completed by Jagan</p></body></html>' > /tmp/sample-app/index.html
[root@ip-10-0-1-114 ~]#
```

echo prints the HTML content.

> writes the content to a file, creating it if necessary or overwriting it if it already exists.

index.html is our sample webpage

```
[root@ip-10-0-1-114 ~]# cat /tmp/sample-app/index.html
<html><body><h1>Welcome to Docker Tomcat</h1><p>Task 1 completed by Jagan</p></body></html>
[root@ip-10-0-1-114 ~]# docker exec tomcat-8080 mkdir -p /usr/local/tomcat/webapps/sample
```

docker exec runs a command inside a running container.

tomcat-8080 identifies the container.

mkdir -p creates the application directory.

/usr/local/tomcat/webapps is Tomcat's web application deployment directory in this image.

sample is the application context name.

```
[root@ip-10-0-1-114 ~]# docker cp /tmp/sample-app/index.html tomcat-8080:/usr/local/tomcat/webapps/sample/index.html
Successfully copied 2.05kB to tomcat-8080:/usr/local/tomcat/webapps/sample/index.html
[root@ip-10-0-1-114 ~]#
```

Copy the html file into the container

docker cp copies files between the host and a container.

/tmp/sample-app/index.html is the source file on your EC2 instance.

tomcat-8080: identifies the destination container.

The remaining path specifies where the file is placed inside Tomcat.

```
[root@ip-10-0-1-114 ~]# docker exec tomcat-8080 ls -l /usr/local/tomcat/webapps/sample/
total 4
-rw-r--r--+ 1 root root 92 Oct 14 10:33 index.html
[root@ip-10-0-1-114 ~]#
```




**Create volume and deploy tomcat container on port 8081.**

```
[root@ip-10-0-1-114 ~]# docker volume create tomcat-data
tomcat-data
[root@ip-10-0-1-114 ~]# docker volume ls
DRIVER    VOLUME NAME
local     tomcat-data
<tomcat-8081 \^J  -p 8081:8080 \^J  -v tomcat-data:/usr/local/tomcat/webapps \^J  tomcat:9.0^C
[root@ip-10-0-1-114 ~]# docker run -d \^J  --name tomcat-8081 \^J  -p 8081:8080 \^J  -v tomcat-data:/usr/local/tomcat/webapps \^J  tomcat>
b40d58c1fd871cd90b56858f741d6b0cd0ce41be686879191397e0d3c3811ddb
[root@ip-10-0-1-114 ~]#
```

-d ---->Runs the container in the background

--name tomcat-8081 -----> Assigns the container name

-p 8081:8080 ------>Maps EC2 port 8081 to container port 8080

-v tomcat-data:/usr/local/tomcat/webapps ------>Mounts the Docker volume at Tomcat's web application directory

tomcat:9.0 ----->Specifies the Tomcat image

```
[root@ip-10-0-1-114 ~]# docker cp /tmp/sample-app/index.html tomcat-8081:/usr/local/tomcat/webapps/sample/index.html
Successfully copied 2.05kB to tomcat-8081:/usr/local/tomcat/webapps/sample/index.html
[root@ip-10-0-1-114 ~]# docker exec tomcat-8081 ls -l /usr/local/tomcat/webapps/sample/
total 4
-rw-r--r--+ 1 root root 92 Oct 14 10:33 index.html
[root@ip-10-0-1-114 ~]#
```




# Welcome to Docker Tomcat

Task 1 completed by Jagan

## Limit the nginx container to 500 MB.

```
[root@ip-10-0-1-114 ~]# docker pull nginx:latest
latest: Pulling from library/nginx
ecc510c1e359: Pull complete
04bdbc7ea6b5: Pull complete
807ab930f3a0: Pull complete
1efc53192927: Pull complete
947d22046304: Pull complete
8020170dc65f: Pull complete
f7e1db57b58b: Pull complete
Digest: sha256:f9ea18bfa4fad859e1ed38259d711da7ccad2c3516e875cec3351a57c859f571
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

```
[root@ip-10-0-1-114 ~]# docker run -d \^J  --name nginx-500mb \^J  -p 8082:80 \^J  --memory=500m \^J  nginx:latest
3e5b2241ed9520fe9701388bb04014180c7c3b55f7698885318c600d1666df90
[root@ip-10-0-1-114 ~]# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                                       NAMES
3e5b2241ed95   nginx:latest   "/docker-entrypoint.…"   10 seconds ago   Up 9 seconds    0.0.0.0:8082->80/tcp, :::8082->80/tcp       nginx-500mb
b40d58c1fd87   tomcat:9.0     "catalina.sh run"        10 minutes ago   Up 10 minutes   0.0.0.0:8081->8080/tcp, :::8081->8080/tcp   tomcat-8081
c10e5635a8c6   tomcat:9.0     "catalina.sh run"        30 minutes ago   Up 30 minutes   0.0.0.0:8080->8080/tcp, :::8080->8080/tcp   tomcat-8080
[root@ip-10-0-1-114 ~]# docker inspect nginx-500mb --format '{{.HostConfig.Memory}}'
524288000
[root@ip-10-0-1-114 ~]# curl -I http://localhost:8082
HTTP/1.1 200 OK
Server: nginx/1.31.6
Date: Wed, 14 Oct 2026 10:59:19 GMT
Content-Type: text/html
Content-Length: 896
```




# Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

*Thank you for using nginx.*

```
[root@ip-10-0-1-114 ~]# docker ps
CONTAINER ID   IMAGE          COMMAND
MES
3e5b2241ed95   nginx:latest   "/docker-entrypoint.…"
inx-500mb
```

Ngninx is restricted with 500 mb

```
[root@ip-10-0-1-114 ~]# mkdir -p /root/nginx-docker
cd /root/nginx-docker
[root@ip-10-0-1-114 nginx-docker]# cat > index.html <<'EOF'
<!DOCTYPE html>
<html>
<head>
    <title>Docker Assignment</title>
</head>
<body>
    <h1>Welcome to Nginx Docker</h1>
    <p>Docker-02 task completed by Jagan</p>
</body>
</html>
EOF
[root@ip-10-0-1-114 nginx-docker]#
```

This creates index.html, the webpage that Nginx will serve.

Next, we'll create the Dockerfile with the correct Amazon Linux package installation commands.

```
[root@ip-10-0-1-114 nginx-docker]# cat > Dockerfile <<'EOF'
FROM amazonlinux:latest

LABEL maintainer="Jagan"

RUN dnf install -y nginx && dnf clean all

COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
EOF
[root@ip-10-0-1-114 nginx-docker]#
```

FROM — selects Amazon Linux as the base image.

LABEL — records the maintainer's name.

RUN — installs Nginx.

COPY — copies your HTML file into Nginx's web directory.

EXPOSE 80 — documents the port Nginx uses.

CMD — starts Nginx in the foreground so the container keeps running.

```
[root@ip-10-0-1-114 nginx-docker]# ls -l
total 8
-rw-r--r--. 1 root root 190 Oct 14 11:55 Dockerfile
-rw-r--r--. 1 root root 180 Oct 14 11:53 index.html
[root@ip-10-0-1-114 nginx-docker]# cat Dockerfile
FROM amazonlinux:latest

LABEL maintainer="Jagan"

RUN dnf install -y nginx && dnf clean all

COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

```
[root@ip-10-0-1-114 nginx-docker]# docker build -t jagan-nginx:v1 .
[+] Building 35.1s (8/8) FINISHED                                                
```

-p 8082:80 -----> Maps EC2 port 8082 to container port 80

--memory=500m ----->Limits the container to 500 MiB

jagan-nginx:v1 ------>Uses your custom image

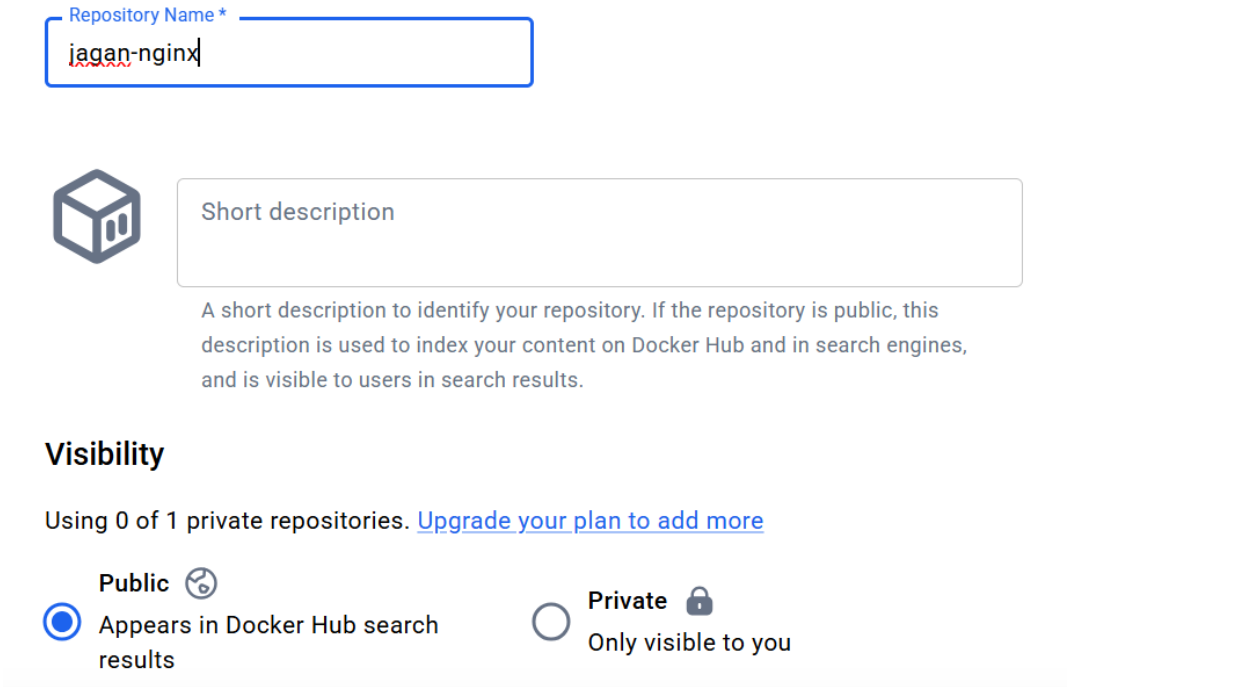
```
[root@ip-10-0-1-114 nginx-docker]# docker ps
CONTAINER ID   IMAGE            COMMAND                  CREATED              STATUS              PORTS
  NAMES
1686c4a466e3   jagan-nginx:v1   "nginx -g 'daemon of…"   About a minute ago   Up About a minute   0.0.0.0:8082->80/tcp, :::8082->80/
  nginx-500mb
b40d58c1fd87   tomcat:9.0       "catalina.sh run"        About an hour ago    Up About an hour    0.0.0.0:8081->8080/tcp, :::8081->
  tomcat-8081
c10e5635a8c6   tomcat:9.0       "catalina.sh run"        2 hours ago          Up 2 hours          0.0.0.0:8080->8080/tcp, :::8080->
  tomcat-8080
[root@ip-10-0-1-114 nginx-docker]# docker inspect nginx-500mb --format '{{.HostConfig.Memory}}'
524288000
```




# Welcome to Nginx Docker

Docker-02 task completed by Jagan

**Push image to Docker hub**



```
[root@ip-10-0-1-114 nginx-docker]# docker tag jagan-nginx:v1 jagan03/jagan-nginx:v1
[root@ip-10-0-1-114 nginx-docker]# docker login
Log in with your Docker ID or email address to push and pull images from Docker Hub. If you don't have a Docker ID, head over to
.docker.com/ to create one.
You can log in with your password or a Personal Access Token (PAT). Using a limited-scope PAT grants better security and is requi
anizations using SSO. Learn more at https://docs.docker.com/go/access-tokens/

Username: jagan03
Password:
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[root@ip-10-0-1-114 nginx-docker]# docker tag jagan-nginx:v1 jagan03/jagan-nginx:v1
docker push jagan03/jagan-nginx:v1
The push refers to repository [docker.io/jagan03/jagan-nginx]
8f0c5392f2b9: Pushed
b8b8f0178da4: Pushed
37ab5dcca5fc: Mounted from library/amazonlinux
v1: digest: sha256:7c3744f98349a8555188990e4dc7e3c9ed26fceb3915a81c1b54749462dd1f66 size: 948
[root@ip-10-0-1-114 nginx-docker]#
```

TAG

v1

Last pushed **1 minute** by jagan03

```
docker pull jagan03/jagan-nginx:v1
```

| Digest | OS/ARCH | Last pull | Compressed size |
|---|---|---|---|
| 7c3744f98349 | linux/amd64 | less than 1 day | 82.74 MB |

Amazon ECR > Private registry > Repositories > Create private repository

## Create private repository

### General settings

**Repository name**

Enter a concise name. Repositories support namespaces, which you can use to group similar repositories.

469132170253.dkr.ecr.us-east-1.amazonaws.com/ jagan-nginx

11 out of 256 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers, and special characters ._-/.

## Private repositories (1)

View push commands | Delete | Actions | Create repository

Search by repository substring

| Repository name | URI | Created at | Tag immutability | Encryption type |
|---|---|---|---|---|
| jagan-nginx | 469132170253.dkr.ecr.us-east-1.amazonaws.com/jagan-nginx | October 14, 2026, 17:54:09 (UTC+05.5) | Mutable | AES-256 |

```
[root@ip-10-0-1-114 ~]# aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 469132170253.dkr.ecr.us
-east-1.amazonaws.com
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[root@ip-10-0-1-114 ~]# docker tag jagan-nginx:v1 469132170253.dkr.ecr.us-east-1.amazonaws.com/jagan-nginx:latest
[root@ip-10-0-1-114 ~]# docker push 469132170253.dkr.ecr.us-east-1.amazonaws.com/jagan-nginx:latest
The push refers to repository [469132170253.dkr.ecr.us-east-1.amazonaws.com/jagan-nginx]
8f0c5392f2b9: Pushed
b8b8f0178da4: Pushed
37ab5dcca5fc: Pushed
latest: digest: sha256:7c3744f98349a8555188990e4dc7e3c9ed26fceb3915a81c1b54749462dd1f66 size: 948
[root@ip-10-0-1-114 ~]#
```

# jagan-nginx

Summary | Images | Lifecycle policy | Permissions | Repository tags

## Images (1) Info

Delete | Copy URI | Details | Scan | View push commands

Filter active images

| Image tags | Type | Created at | Image size (MB) | Image digest | Last pulled at |
|---|---|---|---|---|---|
| latest | Image | October 14, 2026, 18:06:12 (UTC+05.5) | 86.76 | sha256:7... | - |